Neither Memory nor an ODE-RNN: A Pre-Registered Negative Result for Learned Adaptive-Therapy Policies under Burden-Only Observation

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Paper under double-blind review

Abstract

Adaptive therapy tries to keep drug-sensitive tumour cells alive so that they hold resistant cells in check. Reinforcement learning has been used to schedule it, and recurrent or continuous-time policies are natural candidates when the clinician sees only the total tumour burden. We ask whether a policy whose hidden state is an ODE (an ODE-RNN) schedules better than an LSTM or a memoryless network, in a virtual-patient simulator, with the main hypotheses and kill criteria written down before the runs. It does not. Trained by the same algorithm (PQN), a memoryless network that learns a burden threshold matches or beats both recurrent networks across patient ranges, under a parameter shift, and when visits are missed. The task is not trivial, since a per-patient search on the true simulator finds better schedules, but most of that gap does not come from memory, and it is smaller than what training the recurrent networks costs. Cloning a threshold teacher shows that the recurrent networks can represent a good policy and that PQN fine-tuning then damages them, which points to the trainer, not the architecture, as the bottleneck for memory in this setting. Along the way we identify evaluation pitfalls that can turn a spurious difference between architectures into an apparently significant one, and we propose a set of controls for comparisons under partial observation.

1 Introduction

Adaptive therapy treats a tumour so that drug-sensitive cells survive and keep competing with resistant ones, instead of trying to eradicate everything. The idea comes from evolutionary models of resistance, and its first clinical test in prostate cancer paused and restarted the drug according to the PSA level (Zhang et al., 2017). Whether a rule that simple is the best one is an open question, and it is a natural place to ask what a learned policy can add.

Reinforcement learning (RL) has been applied to exactly this. On a Lotka–Volterra (LV) model where the policy sees only total burden, A3C learns a treat-above-threshold rule close to the clinical algorithm (Gallagher et al., 2024). Recurrent policies are reported to help chemotherapy control when part of the state is hidden (Kiram et al., 2026). Policies fitted to patient data are starting to appear as well (Lu et al., 2024). In these studies the architecture of the policy is mostly a fixed choice, and the comparison between architectures is not the question.

That comparison matters because of a common argument. A tumour is a continuous process observed at irregular visits, so a policy whose hidden state is itself an ODE (Rubanova et al., 2019) should carry the right inductive bias, especially when the biology shifts or visits are missed. The evidence for continuous-time networks comes mostly from prediction on irregularly sampled series. To our knowledge there is no controlled test of the argument for control under partial observation, where the policy also has to learn from reward. Adaptive therapy is a small test bed for it, because the simulator lets us compare any policy with the true state.

We run that test for one continuous-time design, the ODE-RNN. The main hypotheses and kill criteria were written down before the runs, so that a null result could not be explained away afterwards; analyses chosen after seeing results are marked as such. The ODE-RNN is compared with a memoryless MLP, an LSTM and an LSTM that is told the gap between visits, all trained with the same algorithm and matched in size. The ODE-RNN never beats the best discrete arm. The obvious objection is that recurrent networks are simply hard to train, so we address it in two ways: a search on the true simulator that measures how much room there is for memory at all, and a warm-start experiment that asks whether the recurrent networks can represent a good policy and lose it in training.

This paper makes three contributions.

1. A controlled comparison. One trainer, one environment and one evaluation protocol, with only the Q-network swapped and the recurrent arms matched in size.

2. A negative result and a partial explanation. There is no ODE-RNN edge in range, under a parameter shift or with missed visits. The headroom split and the warm-start experiment suggest that the recurrent deficit is a learning cost of PQN, with little room for memory to repay it.

3. A record of how the result was reached: four evaluation mistakes that produced a wrong result that looked publishable, the controls that found them, and a table of every registered run.

We do not claim that no memory policy could help, that no other trainer would avoid the deficit (three others did not let a reference beat the memoryless network, so they cannot test it), or anything about real patients; all patients are simulated.

2 Related work

RL for tumour control. Learned dosing policies on tumour ODEs have been reported to approach a clinical threshold rule from burden alone (Gallagher et al., 2024), to approach an optimal-control benchmark when every subpopulation is observed (Kidane et al., 2026), to benefit from recurrence when part of the state is hidden (Kiram et al., 2026), and to prolong time to progression on models fitted to trial patients (Lu et al., 2024). None compares memory with continuous time under one trainer, with uncertainty over training seeds. Optimal control gives a useful contrast: when the drug raises the mutation rate toward resistance, the optimal elimination strategy is a nearly constant intermediate dose, and the gain from closed-loop adjustment is described as marginal (Kuosmanen et al., 2021). A small gain from feedback is consistent with what we find.

Adaptive therapy and its models. The competition between sensitive and resistant cells is modelled with Lotka–Volterra systems, whose parameters can be taken from published ranges (Strobl et al., 2021); the first clinical trial used a PSA threshold rule (Zhang et al., 2017). Our primary regime follows those ranges, and we test on them how much a learned policy can gain over such a rule.

Memory in partially observed RL. Carefully implemented recurrent model-free RL is a strong POMDP baseline; there, as here, an oracle given the hidden state under shared hyperparameters often scored below it (Ni et al., 2022). Across 13 memory models, classical RNNs beat modern ones and a memoryless MLP is often not beaten at all (Morad et al., 2023). Our MLP result is this pattern in an extreme form, and our headroom split (§4) measures how little there was for memory to buy.

Continuous-time networks. ODE-RNNs (Rubanova et al., 2019), neural CDEs (Kidger et al., 2020) and liquid time-constant networks (Hasani et al., 2021) outperform discrete RNNs on irregularly sampled *prediction*. Their advantage is largest when the observation pattern itself is informative (Rubanova et al., 2019; Kidger et al., 2020), and control is named as untested future work (Hasani et al., 2021). We test the control case, with gaps that carry no information.

3 Setup

Virtual patients. (Full protocol: Appendix C.) Sensitive cells S and resistant cells R compete for a shared capacity K:

$$\dot{S} = r_S S\,(1 - \tfrac{S+R}{K})(1 - d_D u) - d_S S, \qquad \dot{R} = r_R R\,(1 - \tfrac{S+R}{K}) - d_R R,$$

with drug effect $u \in [0, 1]$ from one-compartment PK. Patients are Latin-hypercube samples of θ and of the initial resistant fraction f_R and fill level n_0. The *primary regime* follows Table 1 of Strobl et al. (2021): $r_S = 0.027\,\mathrm{d}^{-1}$, $r_R/r_S \in [0.5, 1]$, one turnover $d_S = d_R = d_T$ with $d_T/r_S \in [0, 0.5]$, $d_D = 1.5$, $n_0 \in [0.1, 0.75]$, $f_R \in [10^{-3}, 10^{-1}]$. Fed these parameters and Strobl et al.'s two schedules, our implementation reproduces all 14 of their published times to progression and adaptive-therapy gains that we checked, within 3% (their Fig. 4E and text). A *provisional regime*, on which we first ran the study, has a much stronger cost of resistance (median $r_R/r_S = 0.30$; only 9.8% of its patients fall inside Strobl's sweep ranges). Strobl et al.'s own fits to prostate-cancer patients extend to costs above 50%, paired with low turnover (their Fig. 5C). We therefore report it as a second, high-cost regime, whose ranges are not taken from the literature.

Decision problem. The task is a finite-horizon partially observable MDP (POMDP) with discount $\gamma = 0.99$. The hidden state is (S_t, R_t, C_t, θ): both cell populations, the drug level C_t in the PK compartment, and the patient's parameters. At each of 52 visits the policy chooses a dose $a_t \in \{0, \frac{1}{2}, 1\}$, held for 7 d; the dose is added to C_t and the tumour ODE is integrated to the next visit. Given θ the dynamics are deterministic; the randomness is the sampled patient and the observation noise. The observation is $o_t = \log \frac{S_t+R_t}{S_0+R_0} + \varepsilon_t$ with $\varepsilon_t \sim \mathcal{N}(0, 0.05^2)$, and the policy $\pi(a_t \mid o_{1:t})$ sees only the history of o (the MLP only the latest value). The reward is given below.

Episodes and reward. 52 weekly visits (357 d). The policy sees only log(burden/baseline) with 5% log-normal noise, never S or R, and chooses a dose in $\{0, \frac{1}{2}, 1\}$. Progression (TTP) is a 20% rise over the running nadir, confirmed over 28 d. Training reward per visit is $-\max(b/b_{\min} - 1.2,\, 0) - w \cdot \text{dose}$ (plus a constant), with $w \in \{0.05, 0.1, 0.2, 0.4\}$ as the exposure price.

Tiers. The numbering follows a five-tier robustness plan; Tiers 2 (mechanism shift) and 3 (within-episode drift) are untested (§6). Tier 0 uses the training ranges. Tier 1 scales r_S, r_R by 1.5 and K by 0.6, outside the training box. Tier 4 misses each visit with probability p, i.i.d.; a missed visit gives no observation and no decision, and the last dose continues ("hold").

Arms. MLP (width 441), LSTM (26 units), LSTM+Δt (gap as a second input, the standard discrete baseline for irregular series), a positional MLP (width 386; memoryless, but also sees elapsed time, a standard memory baseline (Morad et al., 2023)), and ODE-RNN (Rubanova et al., 2019). The ODE-RNN is a GRU jump at each observation, with a tanh-bounded latent flow between observations in fixed 7/8 d steps, so gaps of whole weeks compose exactly. All use a LayerNorm Q-readout, trained by PQN (Gallici et al., 2024) with its published recurrent settings, untuned here, except parallelism: 300 updates of 32 patients from a 200-patient training cohort, where the paper uses 1024 environments (S3, in §4, tests 1024).

Statistic. Per patient, $U = \min(\text{TTP}, 357) - w \cdot \text{exposure}$: the TTP endpoint net of the exposure price the policy was trained with. Its mean over patients is a restricted mean TTP net of exposure. The 357-d cap bounds it, so no single patient can dominate it. Where the median of the paired differences changes a reading, we say so. $\Delta(A, B)$ is the mean over w (and over $p \in \{0.2, 0.5\}$ at Tier 4) of the paired mean $U_A - U_B$ on 200 test patients. The 95% CI comes from 2000 bootstrap resamples of *training seeds and patients*. Every cell counts (intention-to-treat); only non-finite rollouts would be excluded, and none occurred. Beside U we report capped median TTP with IQR, median exposure, and the burden integral as a metastasis proxy. Reference rules: no treatment; MTD, full dose throughout; AT50, the adaptive-therapy trial rule; thr0.8, full dose above 0.8× baseline.

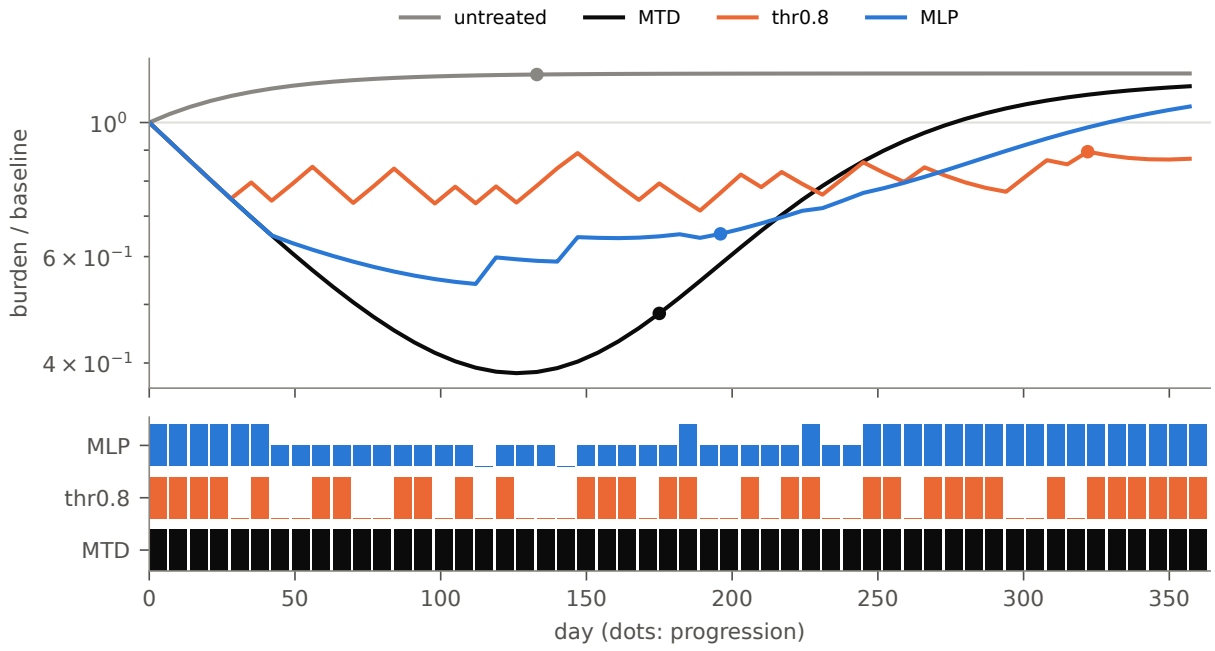


Figure 1: The task. One Tier-1 patient (Strobl ranges) under no treatment, MTD, thr0.8 and a trained MLP (seed 31, $w = 0.1$); dots mark progression. The patient is the one whose drug saved by the MLP against MTD is the median over the patients where it saves any, a fixed rule rather than a hand pick. MTD pushes the burden lowest and progresses at 175 d; thr0.8 and the MLP cut the dose, keep the burden nearer baseline, and progress at 322 d and 196 d. In this model MTD progresses because the drug acts on S only: once S is cleared, the resistant cells R regrow into the freed capacity (competitive release) and the burden follows them upward although the dose stays maximal. The cut-dose policies keep some S alive to hold R back. Bars show the dose chosen each week.

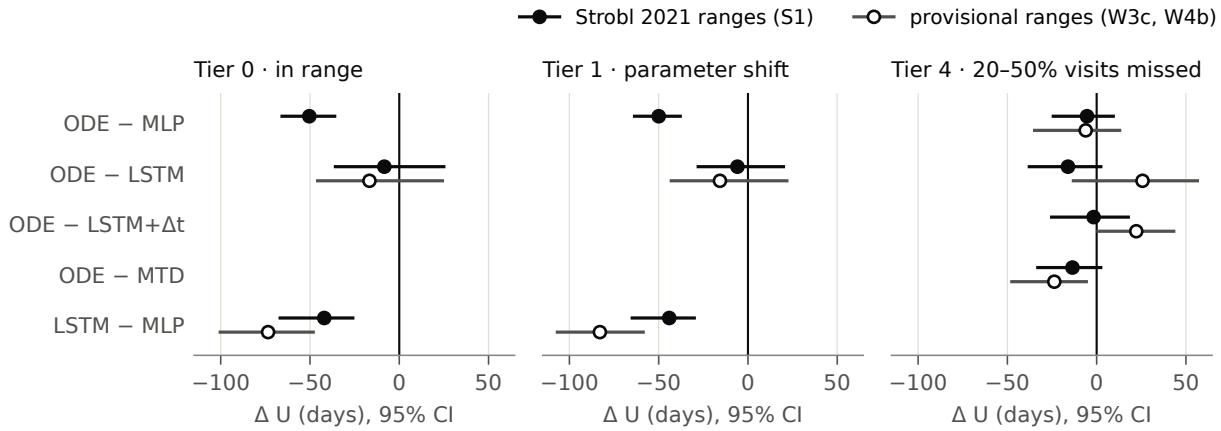


Figure 2: ΔU (days) with 95% CIs over training seeds and patients. Filled: Strobl ranges (10 seeds per tier and arm). Hollow: provisional ranges (Tiers 0/1: 5 seeds, per-cell closed-loop gate; Tier 4: 10 seeds, intention-to-treat). A continuous-time edge needs every ODE row of a panel right of zero.

4 Results

Table 5 lists every run behind these results, with what it changed and whether it was registered before it ran.

Table 1: All paired differences in one place (ΔU, days, 95% CI over training seeds and patients; Strobl ranges, 10 seeds per cell). The ODE-RNN is at or below its discrete counterparts in every column; a clock or memory costs tens of days at regular visits. Rows marked † come from the dropout experiments (V9, V10).

ΔU	Tier 0	Tier 1	Tier 4 ($p = 0.2, 0.5$)
ODE-RNN − MLP	−50.4 [−65.7, −36.2]	−50.0 [−63.5, −37.9]	−5.4 [−24.3, +9.3]
ODE-RNN − LSTM	−8.4 [−35.8, +25.0]	−5.9 [−27.9, +19.9]	−16.1 [−37.7, +2.3]
LSTM − MLP	−42.0 [−66.7, −26.0]	−44.1 [−64.8, −30.1]	—
ODE-RNN − MTD	—	—	−13.5 [−33.0, +2.3]
Positional MLP − MLP	−53.0 [−65.4, −40.9]	−56.8 [−64.2, −48.3]	+2.2 [−13.2, +16.4]
Clock-dropout MLP − MLP†	−7.5 [−10.3, −4.5]	−2.4 [−5.2, +0.4]	—
LSTM, state dropout − LSTM†	−2.2 [−21.0, +17.7]	+4.3 [−13.1, +21.5]	—
ODE-RNN, state dropout − ODE-RNN†	+3.5 [−11.6, +20.3]	+4.2 [−8.8, +18.9]	—

Regular visits: memory does not help. The first question is whether any recurrent or continuous-time arm beats the MLP when visits are regular. On Strobl's ranges (Fig. 2) the ODE-RNN scores −50.0 [−63.5, −37.9] against the MLP at Tier 1 and −50.4 [−65.7, −36.2] at Tier 0, and all ten seeds are negative. Against the LSTM it scores −5.9 [−27.9, +19.9], and the LSTM itself is −44.1 [−64.8, −30.1] against the MLP. Tiers 0 and 1 agree, so the gap is a deficit in control that is already there without any shift. The provisional regime gives the same ordering (LSTM−MLP −82.9 [−106.8, −58.7], ODE−LSTM −15.7 [−43.0, +21.8]).

The deficit is not specific to memory. A positional MLP, which is memoryless but is told the elapsed time, trails the MLP by about as much (−56.8 days at Tier 1, Table 1) and ties the LSTM and the ODE-RNN. Only 0.26 of its doses are explained by the current observation, against 0.48 for the MLP and 0.23 for the ODE-RNN. The trainer seems to take a shortcut (Fig. 3): it fits a schedule keyed to the clock instead of reacting to the tumour. If the clock is hidden in half of the training episodes, the same network recovers (+54.4 [+44.9, +62.3] over the positional MLP at Tier 1) and ties the MLP (−2.4 days). Scrambling the clock after training costs only −14.6 days, with an interval that spans zero, and when visits are missed at random the positional MLP ties the MLP (+2.2 days). The same trick does not rescue the recurrent arms. Resetting their memory in half of the training episodes gains the LSTM and the ODE-RNN about +4 days at Tier 1 (intervals span zero), and both stay 40 to 46 days below the MLP.

The best learned policy is a threshold. Next we ask what the winning policy actually does. A memoryless MLP's dose is a function of the current observation, so what matters is the shape of that function (Fig. 4). On Strobl's ranges the mean dose crosses one half at 0.71× baseline and is full by 0.8×. A one-line rule that doses fully above 0.8× baseline (thr0.8) matches it: mean U is 215.0 for the MLP against 214.9 for the rule at Tier 1, and 282.0 against 282.8 at Tier 0. With PQN's published parallelism (below) the MLP ends slightly above the rule, 289.2 against 282.8 at Tier 0, by dosing less.

What memory could have bought. The third question is how much room the task leaves for memory at all. At regular visits the simulator is deterministic given the doses, so the best policy with full information is just the best dose sequence for each patient, and no trainer is needed to bound it. A search on the true simulator, run per test patient, beats the MLP by +23.1 [+20.3, +25.8] days of U. A threshold rule whose level is tuned to each patient does better still, +26.0 [+23.5, +28.5]: 14 to 19 more days to progression on 46 to 60 fewer days of drug. The task therefore has room that the MLP does not reach.

That room splits into three near-equal parts (Table 2). About a third is a fixed memoryless threshold near baseline, which the MLP could represent but did not learn, because its training reward charges for burden between the nadir line and baseline. Another third comes from seeing the burden exactly instead of with 5% noise, which bounds what filtering the observation history could recover. The last third comes from knowing each patient's best level, which bounds what inferring the patient from history could add. For a threshold policy, then, information that memory could supply is worth at most about 15 days, while the learning cost of the recurrent arms was 40 to 50. This is a split of one rule family's gain and not a bound on memory: the per-patient search (+23.1) is itself below the tuned threshold (+26.0), so it is not the full-information optimum, and the first two rows of the table are post hoc. Read together with the trainer results below, the

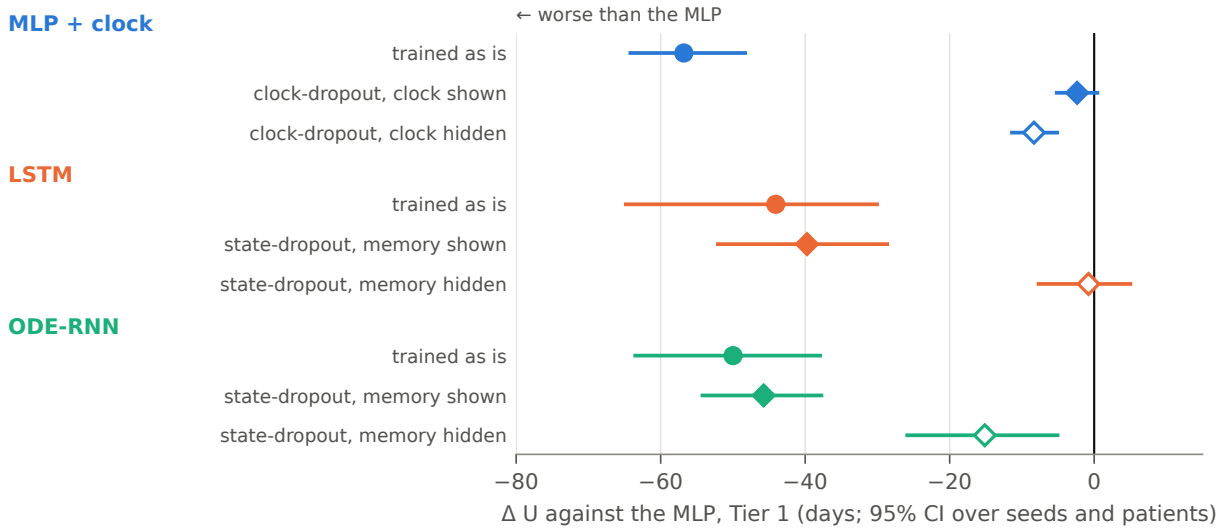


Figure 3: The extra-input deficit is a training shortcut for the clock, not for memory. ΔU against the MLP at Tier 1 (Strobl ranges, seeds 31–40). Circles: trained as is. Filled diamonds: trained with the clock (MLP) or the recurrent state (LSTM, ODE-RNN) unavailable in half of the episodes, tested with it available; hollow diamonds: tested with it hidden. Dropout removes the clock's deficit but not the recurrent ones. Both dropout experiments (V9, V10) were registered before they ran.

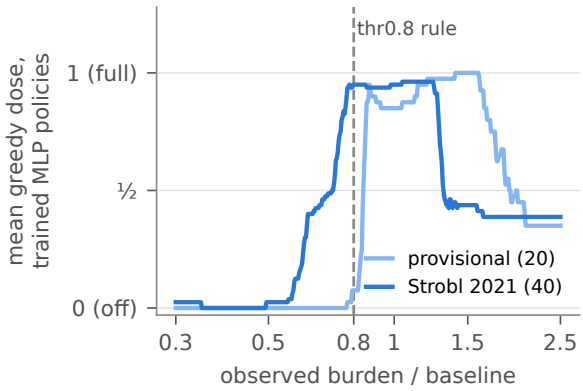


Figure 4: Mean greedy dose of the trained MLP policies against the observed burden. The switch-on point sits near the thr0.8 rule. The step down above ∼1.4× baseline (Strobl ranges) lies where 99.9% of visits are already past progression, so a dose there only costs exposure.

negative result is about PQN at this budget. It is a learning cost larger than the benefit we can measure, which is not a proof that memory has none.

The recurrent arms also reach a lower training reward than the MLP on their own training patients (−20.7 and −23.5 reward units, intervals exclude zero), so they optimise the reward less well. A reference network trained like the arms but also shown S, R and θ never reached the headroom. It scored −63.2 [−75.5, −50.4] below the MLP, then −41.4 [−53.9, −27.6] with PQN's published parallelism, and −19.6 [−25.2, −14.7] with ten times the gradient steps, where it ties the MLP on training reward. That reward has no progression term. Adding one (a survival term paid until progression is confirmed) lifts the reference by +15.4 [+10.2, +20.8] days and brings it level with the MLP (−1.4 [−4.1, +1.2]). It now wins on 62 to 69% of patients (median +2 to +15 days, mostly through less drug), but its mean is lower because of a tail of earlier progressions,

so on the mean it recovers none of the +23 days. The reward was part of the gap, and the rest is PQN's difficulty in using the true state.

Other trainers: none passes our admission rule. The obvious objection is that the deficit belongs to PQN. Our own rule for admitting a new trainer is that a policy shown the true hidden state must beat the burden-only MLP under it. We tried three. PPO (600 updates, untuned, all arms retrained) fails the rule (-1.6 [-10.4, $+5.6$] at Tier 0). So does recurrent Q-learning with an episode replay buffer, a target network and n-step double-Q targets, the recipe of R2D2 (Kapturowski et al., 2019) applied to whole episodes (-3.9 [-8.3, $+0.9$] at Tier 0, -1.8 [-6.0, $+2.2$] at Tier 1). The third is different in kind: evolution strategies, which search the network weights directly on the outcome of whole episodes and use no critic or temporal-difference target (2000 generations of 64 perturbations, -3.1 [-7.0, $+0.7$] at Tier 0, $+1.9$ [-4.1, $+7.1$] at Tier 1). The recurrent arms were not run under the last two. In all three the reference scores no better than the memoryless network, and optimising the episode outcome directly does not change that.

In both of the last two runs the reference ends with a smaller burden integral (about 17 to 20% at Tier 0) at the same median dose, which raised the possibility that the capped-TTP score under-credits it. Scoring the saved runs on the burden integral does not settle in its favour: the smaller burden is paid for with more drug on average (+50 and +22 days of cumulative exposure at Tier 0), because the reference doses heavily in a tail of patients. Neither run can say whether the deficit belongs to PQN, and we do not claim it holds under four trainers. Descriptively under PPO, the LSTM and ODE-RNN again trail the MLP (-40.7 and -42.3 days at Tier 1, and they tie each other), and in 12 and 22 of 40 cells they collapse to a fixed full-dose schedule, the failure described in §5, item 1. We still lack a trainer under which the reference beats the MLP.

Is the LSTM too small? No evidence for it. An LSTM with 64 or 128 units (5 and 22 times the parameters) scores -5.1 [-26.8, $+18.6$] and -3.1 [-25.0, $+26.2$] days against the 26-unit one at Tier 1, and stays -49 and -47 days below the MLP. The intervals are wide, so the registered reading is "inconclusive" against our 10-day gate, but neither size recovers anything. We did not scale the MLP or the ODE-RNN.

Representation or learning? We behaviour-cloned a threshold teacher (a fixed level, or a level tuned to each patient) into each arm. For the fixed threshold every arm reaches the teacher's score: the cloned MLP, LSTM and ODE-RNN get $U = 284.6$, 284.7 and 284.6, so the recurrent networks can represent a good policy. Cloning the patient-tuned level, which a burden-only student can recover only from history, shows no memory gain (LSTM -2.5 [-10.6, $+6.5$]; ODE-RNN $+1.8$ [-9.9, $+13.1$], inconclusive against the 10-day gate). We then fine-tuned the cloned fixed-threshold policies with PQN. The MLP stays where it was (-0.5 [-5.1, $+4.6$]), while the LSTM loses -69.0 [-92.3, -48.1] days and the ODE-RNN -87.7 [-121.1, -60.2], with 2 to 12% of recurrent cells collapsing to open loop. So the recurrent deficit is a cost of training them with PQN, not a limit on what they can represent. It does not show that another trainer would avoid the cost, and the three trainers above cannot either.

Can this trainer use memory at all? Yes, where memory pays. On three POPGym tasks (Morad et al., 2023), run in the JAX port of Lu et al. (2023), the same PQN update trains an LSTM that beats a parameter-matched MLP on two (return, with perfect = +1: RepeatPrevious +1.38 [+1.23, +1.49], RepeatFirst +1.04 [+0.67, +1.39], 10 seeds). On the third, CountRecall, neither arm learns (+0.04). The LSTM is unstable across seeds here as on the cancer task: 4 of 10 RepeatFirst seeds end below +0.1. So the recurrent deficit is not a trainer that cannot represent memory. It is specific to this task and signal. These are two easy tasks, one trainer and no tuning.

Missed visits: no continuous-time edge. Missed visits are where a continuous-time state should help most, so we ask whether it does. Trained with 35% of visits missed, the ODE-RNN is above none of its comparators on Strobl's ranges (Fig. 2): against the MLP -5.4 [-24.3, $+9.3$], against the LSTM -16.1 [-37.7, $+2.3$], against LSTM+Δt -1.7 [-25.2, $+17.7$], and against MTD -13.5 [-33.0, $+2.3$]. It degrades more gracefully than the MLP as visits go missing (+38.9 [+26.7, +48.8] on $U(p) - U(0)$) but less than the LSTM (-20.3 [-33.9, -8.3]), so graceful degradation comes from recurrence and not from continuous time. The ODE-RNN's one advantage over the MLP, at $p = 0.5$, is an exposure artefact: it doses more (median exposure 326 against 194 at $w = 0.1$), and the comparison already nets that drug out in U. The provisional

Table 2: Where the full-information headroom comes from (Tier 0, days of U, mean over the four exposure prices, CIs over seeds and patients). Rows 1 and 2 are post hoc. Fixed levels are chosen on the other nine seeds' patients.

Component	ΔU	Needs
Best fixed threshold under noise ($\approx 1.0\times$) − MLP	+10.8 [+8.1, +13.3]	nothing the MLP lacks
Exact vs noisy observation, fixed threshold	+8.6 [+7.2, +10.5]	filtering (memory)
Patient-tuned vs fixed threshold	+6.6 [+6.0, +7.2]	inferring the patient (memory)
Patient-tuned threshold − MLP (total)	+26.0 [+23.5, +28.5]	

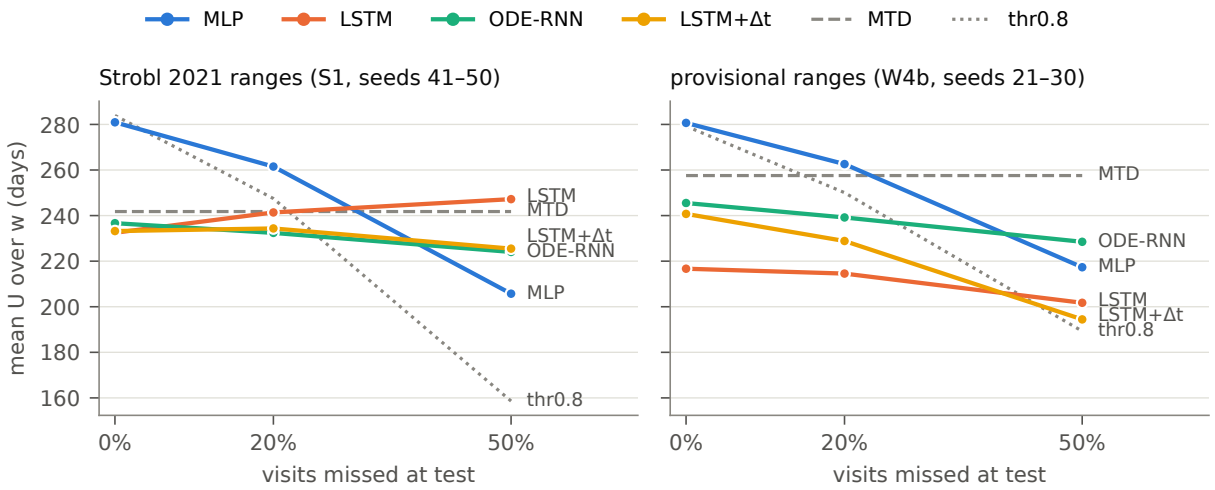


Figure 5: Mean U over the four exposure prices against the share of visits missed at test. All arms were trained with 35% of visits missed; 10 seeds per arm. Grey: MTD (dashed) and the thr0.8 rule (dotted).

regime agrees (ODE−MLP −6.1 [−34.7, +12.9], ODE−MTD −23.7 [−47.5, −5.8]). One Tier 4 effect is unstable: adding Δt to the LSTM changed U by +38.8, +3.5 and −14.4 in three independent runs, so we read differences involving LSTM+Δt as seed noise.

Missed visits remove the value of feedback. The MLP, which is best at regular visits, loses the most as visits go missing (Fig. 5; mean U goes from 280.9 to 205.7 on Strobl's ranges, and for thr0.8 from 283.9 to 158.6). When a visit is missed the last dose continues, so a threshold policy that has switched off keeps the drug off through the gap while the tumour regrows. At 20% missed the MLP is still above MTD (261.5 against 241.8). At 50% no learned policy is above MTD in either regime over our one-year horizon (but see §B for three years). On Strobl's ranges the best arm, the LSTM, ties it (+5.4 [−3.5, +13.6]); the MLP (−36.0 [−44.0, −28.0]) and LSTM+Δt (−16.3 [−29.4, −4.3]) are below it, and the ODE-RNN is at or below it (−17.7 [−39.7, 0.0]). On the provisional ranges all four arms are below it (best: ODE-RNN, −29.1 [−55.8, −8.2]).

These statements are about the mean. For the median patient the LSTM beats MTD (+10.7 [+1.7, +28.0]): about 45% of patients are progression-free under both at one year, and the LSTM uses less drug on them. Replayed unchanged over three years, every learned arm beats MTD on U by 32 to 82 days, but only the LSTM matches its time to progression (+9.0 [−9.4, +28.6] days). The others progress earlier and win on drug saved, as the threshold rules do (§B). The ODE-RNN shows no edge there either (ODE−LSTM −32.1 [−70.9, +9.1]).

Models fitted to patients; monitoring maps. Replaying the saved policies on models fitted to 55 prostate-cancer patients, both negatives replicate (Appendix A). Whether feedback survives missed visits need not depend on architecture at all: with threshold rules only, it stops beating MTD at 10–50% missed visits over one year, and the crossover depends on cost of resistance, turnover and horizon (Appendix B).

Table 3: Controls for claims about memory or continuous time in partially observed RL: what each catches here, and its cost.

Control	Error it catches here	Cost
Memoryless and clock-only baselines	"Memory helps" when a threshold does as well; a clock costs as much as memory (V7)	two extra arms
Trainer-free headroom bound, split by information	"Undertrained recurrent nets": memory could buy ≤ 15 d (O1b)	one search on the simulator
CIs over training seeds and test cases	A patient-only CI that held trained policies fixed (item 3)	≥ 5 seeds per arm
Policy cost next to its score	TTP bought with drug; progressor-only medians (item 4)	one exposure term
Failed runs kept in the count	A validity gate that selected the better cells (item 2); open-loop collapse (item 1)	intention-to-treat reporting

5 Four evaluation errors

Early in the project we made four mistakes that each produced a result that looked publishable. We list them because the controls that found them are cheap and apply to similar claims.

1. The first was open-loop collapse. Under PPO the ODE-RNN gave all 200 patients one identical exposure in 4 of 6 cells, so our first "architecture" result compared feedback with a fixed schedule. The cause was initialisation: each 7-day flow drove the hidden state to a saturated fixed point, and an untrained policy's entropy fell from 1.07 to 0 within one episode.

2. The second was a validity gate that became wrong under missed visits. Once observations were unreliable, near-fixed schedules scored higher than closed-loop cells, so excluding them biased the comparison. We moved to intention-to-treat and added MTD as a reference arm to beat.

3. The third was training instability that looked like an architecture effect. A change of 2×10^{-7} in a float moved one ODE-RNN cell's mean U from 132 to 256 days. Intervals over patients alone ($[-20, -11]$) showed the ODE-RNN worse than the LSTM, and resampling training seeds widened them to $[-43, +22]$.

4. The fourth was a mis-built shifted cohort. Shifting K after computing y_0 put 43 to 50% of Tier-1 patients above capacity. Under full dose 14 to 18% of them diverged to NaN, and these were scored as censored, that is, as successes. A progressor-only TTP median and TTP bought with extra drug misled us in similar ways, which is why every comparison here uses the censoring-aware estimator and the exposure term.

6 Limitations

The study uses one ODE family, Lotka–Volterra with competition between sensitive and resistant cells, and virtual patients only. It makes no claim about clinical outcomes.

The trainer. PQN's settings are the published ones, untuned, except for parallelism (32 patients per update, not 1024). The recurrent arms are unstable in training, and we count that as part of their cost instead of tuning it away. The warm-start experiment supports reading the deficit as a learning cost of PQN (§4), but it uses PQN throughout, so it cannot say whether another trainer avoids the cost. Given the true hidden state, PQN at the registered budgets scores below the burden-only MLP. With ten times the gradient steps it matches the MLP on training reward and still trails on U, and with a progression term in the reward it ties the MLP on mean U. Neither change opened the registered 10-day gate, so we stopped the reward and budget search there instead of tuning until it passed.

The architectures. We test one continuous-time design, the ODE-RNN; liquid time-constant networks (Hasani et al., 2021) and neural CDEs (Kidger et al., 2020) are untested. Missed visits are i.i.d., so the next gap is unknown when the policy doses and carries no information. On a fixed visit schedule with the time to the next visit given as an input, the ODE-RNN is below the MLP on held-out schedules (-21.4 [-42.8, -2.9]). Trained on a pool of schedules it is above the MLP ($+34.8$ [$+16.1$, $+53.7$]), but so is the LSTM, and the MLP never gains from the input ($+1.4$ [-10.3, $+14.4$]), so neither run tests whether a known schedule helps a continuous-time state.

The simulator. Before testing mechanism shift (Tier 2) we checked four other tumour models. Kuznetsov has no resistant population, so MTD is optimal by construction. Hahnfeldt is a mouse model with no resistant population, and our implementation reproduces it in its own units except for two narrow misses (Hahnfeldt et al., 1999). The three-population model behind the clinical adaptive-therapy trial (Zhang et al., 2017) reproduces structurally (22 of 22 equilibria) but not in its published progression times (at most 15 of 40). A tumour–immune model in which chemotherapy also kills immune cells (de Pillis et al., 2006) reproduces its source's qualitative outcomes but leaves no room for a dosing comparison: a rule tuned to each patient beats MTD by only 1.7 days [1.0, 3.0], against our gate of 10, because outcomes are set by the patient's immune state and not by the dosing. LV is therefore the only family here with a resistance mechanism, and every result is LV-only. Within LV the negatives hold on two parameter regimes, under a parameter shift, and on models fitted to 55 patients, and our implementation reproduces Strobl et al.'s published times to progression. Within-episode drift (Tier 3) is untested.

Horizon and metrics. The learned policies were compared over one year only. The monitoring maps show that the value of feedback under missed visits depends on the horizon, so "no learned policy beats MTD at 50% missed" is a one-year statement about the mean. The maps cover threshold rules only, and over one year at low turnover the best rule sits at the grid's most aggressive level. U is a mean. On medians the three-year win holds only where resistance is nearly free, and the sparing rules always carry a larger tumour burden, which our metastasis proxy counts against them.

7 Conclusion

On this task a memoryless burden threshold is a strong baseline, and neither memory nor continuous time beat it, on two parameter regimes and on models fitted to 55 patients' PSA. The task does have headroom, but the part that memory could supply is at most about 15 days, smaller than the learning cost of the recurrent arms (40 to 50 days), and the warm-start experiment points to PQN's training and not to a limit on what the networks can represent. Even given the hidden state, and with a reward that pays until progression, the same trainer only ties the burden-only MLP. When half of the visits are missed, the question changes from which architecture uses feedback best to whether feedback has any value, and over one year, on average, it had none. We think a fair test of a continuous-time policy needs a setting where the gap itself carries information, such as scheduled instead of random visits, and a trainer that can first beat the threshold with the true state.

Broader impact. All patients are simulated; the paper makes no clinical claim and no policy here is a treatment recommendation. We see no significant risk of harm.

Controls for similar claims. Before reporting an architecture effect in partially observed RL, we suggest the five controls of Table 3.

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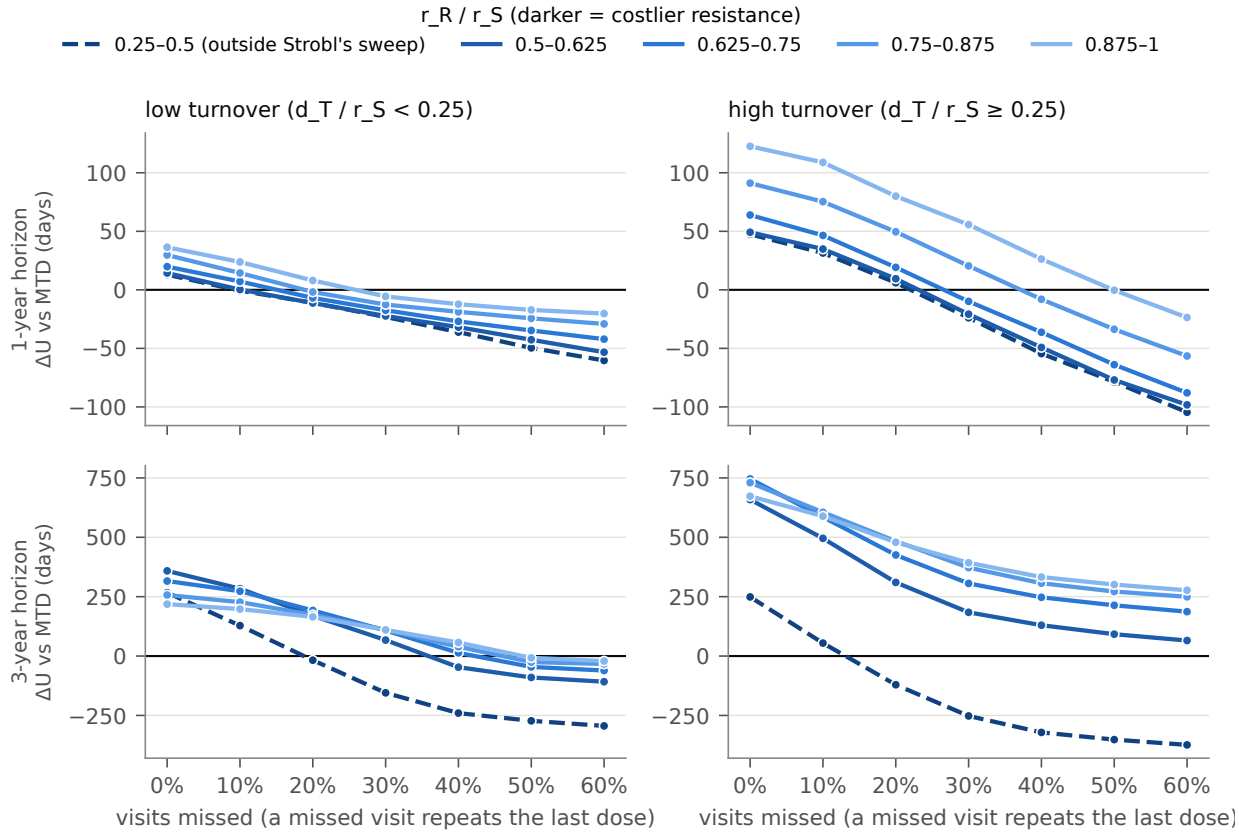


Figure 6: Where monitoring stops being useful. The best threshold rule minus MTD in U (mean over the four exposure prices) on held-out test patients, against the share of visits missed. The threshold is chosen on a separate selection cohort. Lines are bins of r_R/r_S; darker means a costlier resistance, and the dashed bin lies outside Strobl et al.'s sweep. A missed visit repeats the last dose. Rows: the same patients followed for one year (top) and three years (bottom); note the different y-scales. 10 seeds × 200 test patients per bin. Registered: one year (M1) and three years (M1b). The median analysis in the text (M1c) is post hoc.

A Models fitted to patients

Strobl et al. fitted this model to PSA from 67 men in an intermittent androgen-suppression trial (Strobl et al., 2021). We repeated their fit with our simulator: 12 fits fail their $r^2 \geq 0.5$ rule (theirs: 15), and fitted cost and turnover are anticorrelated (-0.73; theirs -0.77). Their relapse separation holds in direction only ($p = 0.04$, post hoc after a mis-registered test). Replaying the saved policies, unchanged, on the 55 kept fits: at regular visits ODE-RNN−MLP is -44.0 [-59.5, -30.5] and ODE-RNN−LSTM -2.4 [-28.1, $+27.7$]; with half the visits missed, no learned arm beats MTD (ODE-RNN -13.2 [-33.2, $+3.3$], LSTM -1.0 [-12.8, $+10.0$]). MTD keeps every fitted patient progression-free for a year, so this replay tests drug use more than progression time. These are models fitted to patients, not patients.

B When does monitoring stop being useful?

Whether feedback survives missed visits need not depend on architecture at all. We binned patients by cost of resistance (r_R/r_S) and turnover (d_T/r_S). In each bin we compared the best burden-threshold rule with MTD, with rules only and no learning (Fig. 6; pre-registered). The rule's level is chosen on a selection cohort and scored on test patients. With reliable monitoring, feedback beats MTD in every bin, by $+13$ to $+123$ days over one year. The share of missed visits at which it stops doing so rises as resistance becomes cheaper: from 0.10 to 0.26 at low turnover, and from 0.23 to 0.50 at high turnover. At 50% missed, the costliest bin

inside Strobl's sweep trails the no-cost end by −25.5 [−31.2, −20.0] days at low turnover and −76.6 [−88.6, −64.7] at high turnover. Feedback thus tolerates unreliable monitoring best where adaptive therapy helps most (Strobl et al., 2021).

The horizon changes the answer. Over one year, MTD reaches the end of the horizon for most patients in the costliest bins, so we followed the same patients for three years (Fig. 6, bottom). The cost effect survives: −82.5 [−96.6, −68.5] and −208.9 [−255.3, −162.3] days at 50% missed. The crossover moves later, to 0.36–0.49 at low turnover and above 0.6 at high turnover. There, sparing thresholds beat MTD by +92 to +301 days at 50% missed. Within three years MTD fails in nearly every patient inside Strobl's range, while sparing dosing holds a minority for the whole horizon. At the no-cost end, 29% of patients never progress under the threshold rule, which uses 72% less drug. Its *median* TTP is nonetheless shorter than MTD's (196 vs 266 days): the gain is a heavy right tail plus drug saved, which U rewards and the median does not. Dosing through missed visits, which helped over one year (+38.5 [+36.1, +41.0]), now helps at low turnover but hurts at high turnover (pooled −5.6 [−13.9, +2.4]).

Medians narrow the claim. U above is a mean over patients. Scored on medians instead (a post-hoc analysis of the same runs, with the threshold also chosen by median), the three-year gain at high turnover and 50% missed survives only in the two lowest-cost bins, $\hat{r}_R \geq 0.75$ (+34.0 [+4.5, +57.7] and +62.3 [+36.9, +90.0] days). In the two costliest bins inside Strobl's sweep it reverses (−137.8 [−166.2, −112.6] and −29.4 [−48.1, −2.0]). The cost effect holds under medians over three years but not over one year at low turnover (−9.9 [−22.5, +7.6]). In all 20 bin×horizon cells at 50% missed, the selected threshold rule has a shorter median TTP and a larger median burden integral than MTD (by 18% at the three-year no-cost end, 4.5× in the costliest Strobl bin). Where sparing wins for the median patient, it wins on drug saved, not on time bought or tumour held down.

C Simulator and protocol in detail

This appendix states every modelling and protocol choice behind the results, so that the experiments can be read without the code.

Tumour model. State (S, R): sensitive and resistant cells, in the equations of §3. Both populations grow logistically toward the shared capacity; the drug acts on the sensitive growth term only, through $(1 - d_D u)$, and both populations die at their own rates d_S, d_R. The resistant population pays a cost of resistance ($r_R < r_S$) and is untouched by the drug, so continued full dosing selects for it, which is what adaptive therapy tries to avoid. The primary parameter ranges are those of Strobl et al. (2021); a patient is one Latin-hypercube draw of (r_R/r_S, d_T/r_S, n_0, f_R) with r_S, d_D fixed. A second, stronger-cost regime is reported separately.

Drug. A dose is a bolus added to a single compartment, cleared exponentially at $0.5\,\mathrm{d}^{-1}$ (closed form between visits). The treatment intensity is $u = C/(C + EC_{50})$ with $EC_{50} = 0.5$, so doubling a dose does not double cell kill. The three doses are $\{0, \frac{1}{2}, 1\}$.

Time. One episode is one year: 52 weekly visits, i.e. 357 d. At every visit the policy sees one observation and chooses a dose; the chosen dose is administered at the visit and the tumour ODE is then integrated for 7 d with a fixed-step Tsitouras 5(4) solver (accurate to about 10^{-4}). An episode always runs the full 52 visits; progression is read off the burden curve afterwards. The ODE-RNN advances its latent state through each 7-day gap in 8 steps of 7/8 d; the MLP and LSTM treat one visit as one step. At Tier 4 a missed visit produces no observation and no decision, the previous dose continues, and the discrete-time arms receive no information about the gap except the LSTM+Δt arm.

Observation. The policy sees one number per visit, log(burden/baseline), where burden is $S + R$ times a log-normal noise factor with 5% standard deviation of the log and baseline is the same patient's burden at visit 0. It never sees S, R, θ or the visit index (except the positional MLP, which sees elapsed time).

Reward and score. The training reward at a visit is $-\max(b/b_{\min} - 1.2,\, 0) - w \cdot \text{dose}$ plus a constant, where $b_{\min}$ is the lowest burden so far; the 1.2 matches the progression definition. One network is trained per exposure price $w \in \{0.05, 0.1, 0.2, 0.4\}$ and seed. The reported score $U = \min(\text{TTP}, 357) - w \cdot \text{exposure}$ uses the same w. Exposure is the cumulative dose over the episode. TTP is the first time the burden has stayed 20% above its running nadir for 28 consecutive days (censored at 357 d if it never does).

Cohorts and seeds. For each seed, 200 training patients are drawn and 200 held-out test patients from the same ranges (Tier 0). Tier 1 rescales r_S, r_R by 1.5 and K by 0.6 and recomputes the initial state so that no patient starts above capacity (an earlier build did not; §5). Each (arm, w, seed) cell is trained from scratch on that seed's training cohort and then scored greedily on its test cohort. The primary regime uses 10 seeds per arm; the provisional regime uses 5.

Training. Every arm is trained by PQN with the settings of Table 4: 300 updates, each collecting 32 training patients' full episodes with ε-greedy exploration, followed by 4 epochs of 4 minibatches of whole patient episodes (never shuffled transitions, because a recurrent Q-value depends on the visits before it). Learning rate, discount and λ are the published values and are the same for every architecture. No arm is tuned.

Comparisons. For two arms A, B the paired difference is $U_A - U_B$ per test patient, averaged over w (and over missed-visit rates at Tier 4), and the 95% interval is the 2.5th and 97.5th percentiles of 2000 bootstrap means that resample training seeds and patients. The registered gates and readings ("edge", "deficit", "ties") and every run's pass and kill criteria were written before the run (Table 5).

What the simulator cannot show. One resistance mechanism (competition), parameters from published ranges and not individual patients, one drug, weekly observations of total burden only, and a one-year horizon in the primary results. Section 6 states the consequences.

D Reproducibility

Table 4: Settings shared by every arm. PQN values are the published recurrent settings (Gallici et al., 2024), untuned, except parallelism.

Trainer	PQN, Peng's Q(λ), no replay, no target network
Learning rate	3×10^{-4}, linear decay to 0
γ, λ	0.99, 0.5
Epochs × minibatches	4×4, minibatches of whole patient episodes
ε-greedy	$1.0 \to 0.005$ over the first 10% of updates
Max gradient norm	0.5
Budget	300 updates × 32 patients × 52 visits (S3: 1024 patients)
Reward scale	0.01
MLP	one hidden layer, width 441, LayerNorm
LSTM, LSTM+Δt	26 units, LayerNorm Q-readout (Δt in units of 7 d)
ODE-RNN	GRU jump, 16-d latent, field width 64, fixed 7/8 d steps
PK	bolus, clearance $0.5\,\text{d}^{-1}$, $E_{\max}$ effect, $EC_{50} = 0.5$
Cohorts	200 training, 200 test patients (Latin hypercube)
Seeds	10 per arm and tier (primary regime); 5 (provisional, Tiers 0/1)

Everything is written in JAX, Equinox and Diffrax and runs on CPU (one Intel i7-13800H); rollouts use fixed-step solvers, which we found accurate to $\sim 10^{-4}$ and faster than adaptive stepping or GPU at this batch size. The largest single run (V2, 80 cells at 1024 patients and 3000 updates) took about 150 CPU-hours. Code, pre-registrations and the dated research log referenced in Table 5 will be released with the paper.

E Registered runs

Run names and log lines refer to the dated, append-only research log released with the code; each run's hypothesis and pass/kill criteria were registered before it ran.

Table 5: Every registered policy comparison, in order, including void and invalid ones. "Valid" means both arms passed the validity rule registered for that run. "Reg.": yes = registered before it ran; post hoc = chosen after earlier results (O1b only partly). Run names used in the text index this table.

Run	Change	Outcome	Reg.	Log entry
Phase 2	MLP vs ODE-RNN, PPO, Tier 1	void: cohort above capacity	yes	L242
Phase 2 rerun	cohort fixed	FAIL; ODE open-loop in 4/6 cells	yes	L275, L312
Phase 2b	PPO fixes + closed-loop gate	invalid: 3/6 ODE cells open-loop	yes	L391
Phase 2c	PQN for both arms	FAIL, first valid run	yes	L424
W3	+ LSTM, exposure frontier	invalid: 1 open-loop cell per arm	yes	L496
W3b	fresh seeds, restricted-mean TTP	invalid: same	yes	L526
W3c	per-cell gate, U primary	valid; no memory or CT edge	yes	L565, L596
W4	Tier 4, trained at 35% missed	invalid: gate removed better cells	yes	L632
W4b	intention-to-treat, MTD reference	FAIL: no CT edge	yes	L668
S1	Strobl ranges, W3c + W4b designs	FAIL: no CT edge, Tiers 0/1/4	yes	L766
S2	reference also sees S, R, θ	below the MLP	yes	L804
S3	S2 at PQN's published parallelism	still trainer-limited	yes	L840
M1	threshold rules vs MTD, cost × turnover bins	met: $p^\star$ 0.10–0.50	yes	L896
M1b	M1 over three years	met; crossover moves later	yes	L932
M1c	M1, M1b on medians (post hoc)	mixed: holds only if $\hat{r}_R \geq 0.75$	post hoc	L968
W5-pre	rules on Hahnfeldt, Kuznetsov	FAIL: no TTP range; Tier 2 cut	yes	L1004
T1	S3 with 10× gradient steps	reward-limited	yes	L1042
V2	T1 + survival until progression	trainer-limited; ties MLP	yes	L1344
R2	Strobl's fits to the Bruchovsky cohort	2/3 reproduce; relapse weaker	yes	L1075
P1	saved policies on the 55 fits	met: both negatives replicate	yes	L1106
V5	reproduce Hahnfeldt, Zhang	FAIL: 3/5, 0/4 targets	yes	L1176
V5d/e	Zhang 2017 + SI	structure ok; times ≤15/40	yes	L1216–48
O1	full-information search, no trainer	headroom +23 d	yes	L1279
O1b	split of O1's headroom	≤15 d of a threshold family's gain needs memory	post hoc	L1310
V7	positional MLP on S1's design	hurts at Tier 1; ties at Tier 4	yes	L1414
V1	reward, medians, 3 y (post hoc)	no misalignment; LSTM > MTD (medians)	post hoc	L1136
V8	clock scrambled at test, V7's policies	scrambling costs 14.6 d, CI spans 0	yes	L1437
W8	scheduled visits + lookahead input	check failed (+9.7 [−0.4, +19.4]); nothing closed	yes	L1457
V9	clock-dropout MLP	deficit removed (+54.4 vs positional MLP)	yes	L1477
W9	W8 on a pool of schedules	gate failed; ODE = LSTM	yes	L1498
V10	state-dropout LSTM, ODE-RNN	deficit remains (+4 d, CI spans 0)	yes	L1518
V11	LSTM at 64, 128 units	inconclusive vs gate; still −47 to −49 d vs MLP	yes	L1558
W13	rules on de Pillis 2006 (chemo kills immune cells)	reproduced (7/7, amended); headroom FAIL: best rule − MTD +1.7 d vs gate 10	yes	L1633
W11	clone threshold teachers, then PQN fine-tune	clones match teacher; fine-tune costs LSTM −69, ODE-RNN −88 d, MLP −0.5	yes	L1597
V12	S1 design under PPO	gate failed: uninformative about trainers	yes	L1578
V13	reference and MLP under episodic-replay recurrent Q-learning	gate failed (−3.9 [−8.3, +0.9]); recurrent arms not run	yes	L1651
V14	reference and MLP under evolution strategies on the episode outcome	gate failed (−3.1 [−7.0, +0.7]); recurrent arms not run	yes	L1668
V15	V13/V14 gate re-scored on the burden integral	failed: smaller burden, but more drug (+50, +22 d)	yes	L1685
W10	PQN, LSTM vs MLP on POPGym memory tasks	met: headroom on 2 of 3 tasks	yes	L1540